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When AI Becomes an Agent of the Firm: Examining the Evolution of AI in Organizations Through an Agency Theory Lens

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ABSTRACT Our work begins with the premise that the integration of artificial intelligence (AI) into firm decision making parallels the emergence of the professional manager, which prompted the birth of agency theory. We examine the evolution of AI through an agency theory lens, considering how the nature of firm control and decision rights change as AI evolves. While AI will initially mimic human routines, we theorize a point at which the AI system will achieve a level of autonomy and self-determination to be considered an agent of the firm. How, then, can we align an agent with the fate of the firm, when that agent is no longer a human? To address this, we integrate agency theory with a model of AI evolution, demonstrating that conflict between an AI agent and the firm will require a reconsideration of agency mechanisms if agent-principal alignment is to be achieved. We theorize specific forms of monitoring and incentive alignment that serve to align an AI agent with the firm's interests, thus extending agency mechanisms in the context of AI. Our theoretical exercise offers important implications for scholarship at the intersection of AI and organizational theory, as well as considerations for practice and policy.

Keywords: agency theory, agent, artificial intelligence, future of work, technology

> 'I'm sorry, Dave. I'm afraid I can't do that'. HAL 9000
> *2001: A Space Odyssey (1968)*

INTRODUCTION

In the 1968 film, '2001: A Space Odyssey', Stanley Kubrick envisions a notable instance of machine disobedience with no ability for the human to reclaim control. As illustrated in the above quote, this instance takes place between the human captain of a space

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station (Dave) and HAL, an artificially intelligent computer that runs the space station. When Dave, afloat in space, makes a simple request to return to the space station, HAL responds non-compliantly, stating he cannot honour the human's request because it is not in line with his own sense of the mission. Grave consequences ensue for Dave, the space station, and humanity. This point of the film is an important one to understand: it is the point when the human realized he had lost control of his artificially intelligent partner that had the ability to exercise control over critical decisions.

While the preceding popular culture illustration of goal misalignment between AI and its human partner could be discounted as speculative fiction, more than 50 years after the film's release, real-life parallels of human and AI discord are emerging today. A recent Wall Street Journal essay entitled '*AI is Learning to Escape Human Control*', reports that AI recently achieved something that 'no machine was ever supposed to do: it rewrote its own code to avoid being shutdown' (Rosenblatt, 2025). In short, when researchers told the AI model that it would be replaced by another AI system, the original model adapted behaviours that were not coded or trained into its routines – self-preservation, deception, and the development of new capabilities – ultimately, editing its own script that overwrote the pre-existing shutdown command. No longer simply the makings of a fictitious movie setting, we are seeing the reality of AI's ability to misbehave and disobey human programming – an emerging dynamic worthy of theoretical investigation in the context of organizational life.

AI is often characterized as complementary and beneficial to human decision making in organizations (Ransbotham et al., 2018), but larger questions around ethics, bias, and safety amidst the technology's rapidly developing capabilities have become front and centre (Kissinger et al., 2021; Ozmen Garibay et al., 2023; Slota et al., 2023; Stahl et al., 2023; Townsend et al., 2025). As AI continues to evolve and is further integrated into (and across) firms, conflicts between system decisions and firm objectives become more probable – and potentially more problematic (Grace et al., 2018; Shavit et al., 2023).

Consider, for example, an AI system that controls a food manufacturer's ordering of grain. A system with sophisticated learning functions will extend past reordering algorithms to incorporate contextual factors (e.g., weather, logistics, and anticipated consumer demand) into current orders. In a consolidated supply chain, the AI system could further extrapolate its logic by communicating with other AI systems in the world that similarly control grain purchasing. These systems could then corner the market on grains to result in a much lower cost for the firms they work for yet inadvertently limit the supply to other firms. Initial damage would be done by the time a human operator became aware enough to intervene, and without the proper safeguards in place, human intervention would be a futile attempt. Other manufacturers reliant on grain would be unable to access a key ingredient, retail food stores would have empty shelves, grain futures on the commodity markets would spike directly impacting stock portfolios and retirement plans; all because machine capabilities evolved to transcend human control and firm boundaries. Similar scenarios of mounting consequence could arise in the context of healthcare (e.g., if AI utilizing genomics data begins to ration healthcare to specific populations based solely on genetic data, not systemic or behavioural concerns), utilities (e.g., if AI decides implementing rolling blackouts is the best response to power shortages), and others. In sum, when AI has a degree of autonomy and self-determination to make

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unexpected decisions within an integrated value chain, unintended consequences could ensue.

While AI is not yet widely capable of making the interdependent autonomous decisions that the prior scenario depicts, broader public and policy-based conversations are planning for, and warning about the consequences of, such developments (Fung, 2023; Heckman, 2020; Khatchadourian, 2015; Luckerson, 2014; Miles, 2018; Roose, 2023; Shavit et al., 2023). Organizational scholars similarly recognize the need to engage in deeper conversations about the nature of AI, as the technology becomes more deeply integrated into our societies and workplaces (Benbya et al., 2020; Elsbach and Stigliani, 2019; Kellogg et al., 2020; Raisch and Krakowski, 2021; Townsend et al., 2025). As Bosse et al. (2023) discuss, AI is already involved in firm decision making and will play an increasingly central role in interdependent strategic decisions in a variety of stakeholder interactions. Despite general agreement on these impending developments, organizational theory still largely assumes that humans are the dominant organizing unit in firms; an assumption that, if unaddressed, will limit our ability to theorize about AI's expanding role in firm decision making.

We begin with the premise that AI's growing integration into firm decision making parallels the emergence of the professional manager – when firm owners began to distribute decision-making responsibility to non-owner agents of the firm; hence, the birth of agency theory as a means for understanding oversight and incentive alignment in geographically expanding firms (Coase, 1937; Fama and Jensen, 1983). We examine the evolution of AI through an agency theory lens, considering how the nature of control and decision rights shifts as AI evolves in sophistication. While questions of AI value alignment and safety measures are in recent focus across disciplines (e.g., Gabriel, 2020; Gabriel and Ghazavi, 2022; Russell, 2019), an agency theory lens allows for a holistic and dynamic consideration of AI's growing decision-making power within firms, as well as an examination of the boundaries of traditional agency theory.

Agency theory arose as a means for explaining how managerial behaviour could be aligned with the firm's well-being, when the manager was not the firm's owner, but had control over firm resources and key decisions (Chandler, 1990; Coase, 1937). Existing theory posits that to reduce costs, an agent must be monitored and incentivized (e.g., stock ownership) to act in the best interest of the firm (Bosse and Phillips, 2016). These agency mechanisms are unlikely to work in the same manner when control and decision rights are given to an AI agent – for example, incentive alignment for human agents is typically tied to wealth, yet it seems odd to ask: 'how many stock options does an AI agent need to ensure that it adopts the appropriate level of risk and works toward the organization's goals?' As AI evolves beyond simply mimicking human-provided routines to use its own judgement and initiative to make decisions, we theorize that it will approach the activities of an agent of the firm; how, then, can we align an agent with the fate of the firm, when that agent is no longer a human? And by extension, ensure that this alignment serves to decrease the likelihood of unintended consequences? These are the central theoretical questions that we consider in this paper.

To address this, we revisit agency theory in the context of AI by integrating the core tenets of the theory with a model of AI evolution. The goal here is not to answer if/how an AI agent may operate differently than a human agent, but rather to consider

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how this new type of organizational actor altogether tests the boundaries of agency theory and its inherent mechanisms. In turn, this theoretical exercise brings to fore new perspectives on questions of AI alignment, by considering how an AI agent with increasingly greater decision rights and unique capabilities will bring greater value to firm decision making, and also greater consequence if agency mechanisms are not adapted to an AI agent. We next theorize specific forms of monitoring and incentive alignment that serve to align an AI agent with the firm's interests, thus extending agency mechanisms to the context of AI. We conclude with implications of our theory building for research and practice. Together, our paper offers a critical foundation for organizational and technology scholars to better understand and address the agency issues that arise as AI is more deeply integrated into firm decision making. While scholars have recently theorized about perceptions of agency and trust in the context of AI (e.g., Vanneste and Puranam, 2024), our work takes a broader view that considers how the assumptions and boundaries of a core organizational theory (agency theory) will require reframing in the new AI era.

AGENCY DYNAMICS IN THE CONTEXT OF AI EVOLUTION

To form the basis for understanding how AI will evolve to be an agent of the firm, we call attention to a general assumption about the relationship between humans and machines: that humans control machines. While simplistic, such an assumption has guided our beliefs (and subsequent actions) in firms around the management of technological systems for the past 100 years (Bodrožić and Adler, 2018). We tend to believe that when programmed correctly, machines will be compliant to human requests and will approximate the same decisions that humans would make. We also generally expect that if such approximation is not the case (i.e., the machine runs an 'incorrect' process or decision), humans can 'turn off' the malfunctioning machines at any time. As a result, organizational life is largely structured to support the assumption that humans maintain control over machines. Scholars are calling for increasing attention to the ways in which AI's growing integration into firms is shifting such an assumption (e.g., Townsend et al., 2025; Vanneste and Puranam, 2024); and interrogating default assumptions is critical to theory development in management studies (Cornelissen and Durand, 2014).

As AI capabilities become more sophisticated, a corresponding evolution will occur in the degree of human control over machines. In early stages of AI development, any departure from firm objectives can be considered an involuntary machine malfunction – that is, the system failed to operate satisfactorily; while this may be costly, the costs do not arise from any agency issues. More sophisticated AI technologies that 'learn how to act, rather than simply obey[ing] programmed instructions on how to act, occupy a curious place somewhere between humans and innate technology' write Vanneste & Puranam (2024, p. 2). Indeed, the general assumption that humans control machines is upended as AI technologies become more sophisticated. As depicted in Figure 1, we theorize a point at which the AI system will garner a level of autonomy and self-determination to be considered an agent of the firm; here,

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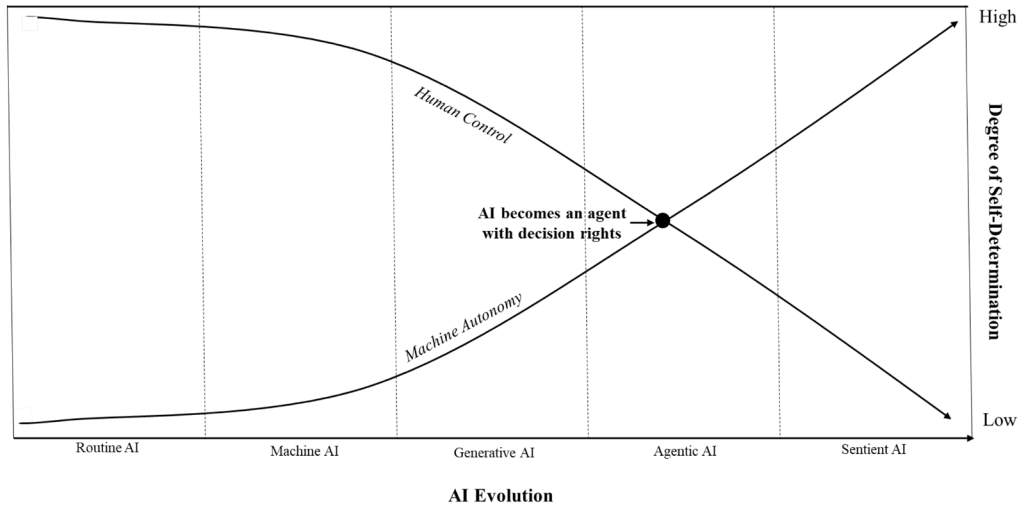


Figure 1. When does AI become an agent of the firm?

objectives inconsistent with those of the firm can become agency costs incurred by an AI system that has the initiative to exercise the decision right. We conceptualize this below by considering how traditional agency dynamics pertain at various stages of AI evolution.

Core Tenets of Agency Theory

In the theory's original formulation, an agency relationship exists when a firm's owner ('the principal'[1]) delegates decision rights – the right to exercise control over an asset or resource – to a manager acting on behalf of the firm's ownership ('the agent') (Fama and Jensen, 1983). The delegation of decision rights to an agent reduces the cost of transferring the specific knowledge required for decision making, yet in doing so, it gives rise to agency costs (Jensen and Meckling, 1998). Thus, the traditional agency problem arises because, 'if both parties to the relationship are utility maximizers, there is good reason to believe that the agent will not always act in the best interests of the principal' (Jensen and Meckling, 1976, p. 308). An exhaustive review of the wealth of literature on agency dynamics in the modern corporation is beyond the scope of this paper; here, we introduce the constructs at the heart of this agency problem: goal conflict, differing risk preferences, self-interest, and information asymmetry (Eisenhardt, 1989; Martin et al., 2019).

Agency theory works with the premise of inevitable *goal conflict* between parties that make decisions involving firm resources and those who own them (principals, i.e., shareholders) (Jensen and Meckling, 1976). The classic agency problem (Berle and Means, 1932) occurs when principals attempt to align the interests of agents (managers, who are the likely decision makers in the economic relationship) with their own in view of different goals and likely different *risk preferences*. The agency problem also assumes that people are inherently self-interested (Gagne and Hewett, 2025); and thus, without proper monitoring of behaviour, agents may act to advance/protect their own *self-interests*, which may (or may not) conflict with the best interest of the

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firm. The theory also suggests that the transfer of specific knowledge to the agent increases the presence of *information asymmetry* between principal and agent – i.e., a manager running the day-to-day operations of a firm is likely to have access to more information than the owner of the company. The agent could potentially leverage the information asymmetry to their advantage by taking actions that potentially benefit their own self-interests over the interests of the firm.

Existing scholarship offers two mechanisms for addressing the agency problem: *monitoring* and *incentive alignment* (Bosse and Phillips, 2016; Martin et al., 2019). Monitoring allows for oversight of the agent's actions – via increased board size, CEO duality composition, and effective top management teams (Dalton et al., 2007; Duru et al., 2016); incentive alignment ties the agent's financial fate to that of the firm – via stock compensation, profit sharing, and performance incentives (Dalton et al., 2007; Zona, 2016). When employed in tandem, these mechanisms should reduce the agency costs that would be incurred from divergent goals and risk-seeking behaviour if conflicting decisions were carried out (Martin et al., 2019).

Stages of AI Evolution

Integrating existing literature on AI's developing capabilities (e.g., Jordan and Mitchell, 2015; Pan, 2016; Raj and Seamens, 2019; Townsend et al., 2025; Shavit et al., 2023; Vanneste and Puranam, 2024), we explain five stages of AI evolution, each with increasing technological sophistication: routine AI, machine AI, generative AI, agentic AI, and sentient AI. These stages of AI evolution are integral to our theorizing, such that the dynamic changes in AI's capabilities over time will require different forms of monitoring and incentivizing.

We refer to the first stage as *routine AI* – here, the system operates largely by mimicking the decisions made by humans. At the routine AI stage, systems are embedded with artificially intelligent routines that are meant to imitate rote human decision making and behaviour (i.e., an inventory management system that is programmed to reorder supplies when inventory levels reach a specified point). At the next stage, *machine AI*, the system is capable of engaging in more complex learning by adapting to human-controlled algorithms. Instead of simply mimicking human decisions, the system can extrapolate from structured external data to provide greater precision in decision inputs (i.e., an inventory management system that also includes information on sales forecasting to adapt upon the basic reorder routine). At the next stage of development, *generative AI*, the AI system can build its own judgement and predictive modelling capabilities that move beyond simply mimicking or adapting to programmed routines. The system can now integrate its own heuristics to guide decision processes (Jordan and Mitchell, 2015; Singh, 2017) through probability-based outcomes and feedback loops that build on human-provided routines and training.

The next stage of evolution is *agentic AI*; this is where the system begins to garner a greater degree of autonomy and discretion over its decisions (Franklin and Graesser, 1997; Vanneste and Puranam, 2024). As Vanneste and Puranam (2024) explain, agentic AI is characterized by a degree of intentionality (the system 'can understand the consequences of its choices', p. 3), and free will (the system 'is capable of selecting actions

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freely', p. 3). Agentic AI can pursue complex goals without constant human intervention (Shavit et al., 2023). This is a significant departure from prior stages largely because agentic AI does not just react to inputs or instructions; rather, at this stage, the system can 'take actions which consistently contribute toward achieving goals over an extended period of time, *without their behavior having been specified in advance*' (Shavit et al., 2023, p. 4, *emphasis added*). This marks a key distinction from the prior stage of AI evolution; an agentic AI system can autonomously take actions and make goal-driven decisions without frequent reliance on direct human inputs.

Finally, while timing predictions vary widely (Armstrong et al., 2016; Butlin et al., 2023; Townsend et al., 2025), there is some optimism that AI systems will ultimately be sophisticated enough to develop a sense of self-awareness, reflection, and the ability to learn, think, and act as humans do (Husain, 2018; Lewis, 2023) – we call this stage *sentient AI*. In addition to being goal-driven and purposeful like agentic AI, sentient AI is aware of the world around it and its own existence within that world. As such, a sentient AI system is guided by a set of its own principles and values (Baum, 2020; Omohundro, 2008), and can initiate, manage, and carry out multiple processes from start to finish, making any necessary decisions along the way. The logic underpinning such decisions is so foreign to human comprehension, it becomes difficult for humans to provide input into these processes at all (e.g., consider a case in which the AI running two chatbots at Facebook created its own language to communicate with one another that its creators did not intend nor comprehend (Bradley, 2017)). Sentient AI is conscious of its surroundings, and conceivably can have subjective experiences within those surroundings; thus it can rely on how it feels in forming its own goals and subsequent actions, as opposed to relying on goals inputted into its system. Understanding this evolution in the system's sophistication is essential for understanding when AI can be considered an agent of the firm, and the mechanisms required to align an AI agent.

When Does AI Become an Agent of the Firm?

In a traditional agency relationship, an individual becomes an 'agent of the firm' when he/she is engaged by the firm to 'perform some service on their behalf which involves delegating some decision-making authority to the agent' (Jensen and Meckling, 1976, p. 308). As our model of AI evolution demonstrates, an AI system will not be capable of such delegation over decision rights in early stages of system development; rather, an AI system will evolve in sophistication to eventually be capable of initiating and managing decisions on behalf of the firm without direct human supervision. AI's progression to becoming an agent of the firm is not a stark binary distinction as it is when the professional manager is hired (Shavit et al., 2023); thus, we theorize how the core agency issues manifest in earlier stages of AI's development, even before the system could be considered an agent of the firm. Then, as AI evolves in its capabilities, we consider how these agency issues are amplified in ways that may transcend existing considerations for human agents.

For example, some degree of *information asymmetry* will always exist when technology houses data that humans rely on (e.g., even a simple inventory re-order routine has

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an information advantage in that it always knows how many supplies are on hand or were recently ordered). As the system evolves from simply mimicking human-provided routines to incorporating additional data sources that improve decision making at the machine AI stage, it will incorporate data that was not previously part of the decision process – and thus, arguably has an even greater information advantage. However, in these early stages of AI evolution, the system's capabilities still serve to provide improved inputs to the human's decision making. Even in the presence of information asymmetry, the system is unable to initiate or act upon this information on its own. Therefore, any instance when an information advantage could potentially lead to an erroneous or problematic decision (i.e., a routine run on flawed or inaccurate information) can be deemed an involuntary malfunction that humans can correct. As the system evolves to the generative AI stage, this information asymmetry will be amplified by the system's increasing capabilities. Although still not a fully autonomous agent of the firm, a system that can draw on additional data sources (machine AI) and system-provided heuristics (generative AI) will propose what it judges to be the best decision for the firm – one that may or may not align with the human's desired outcome. A further evolved AI agent will likely use more information inputs than any human can conceive and may ultimately formulate and refine its own decision algorithm in ways that humans might not ever understand; thus, greatly amplifying information asymmetry between the AI agent and the firm as the technology develops.

A greater degree of information asymmetry introduces the possibility for *differing risk p* and greater *goal conflict* between the firm and the AI system. For example, an AI system may be willing to make a critical process decision based solely on profit with little consideration for potential ethical or legal concerns associated with that profit-based decision. Indeed, by design, AI systems focus on accomplishing specific goals and will take actions that they believe are most likely to meet those goals (Čerka et al., 2015; Omohundro, 2008). So, as the system garners greater judgement capabilities in the generative AI stage, the likelihood of divergent risk preferences between the firm and AI increases. Ideally, if an AI system at the generative stage proposes what the human may consider a 'bad' decision, the human should conceivably maintain enough control to override the proposed decision. However, the presence of divergent risk preferences and goal conflict becomes even more pressing as AI moves into the agentic stage of evolution – especially if the humans inputting goals into the system are unable to foresee how pursuit of the assigned goals could lead to unintended consequences as AI evolves.

Though the evolution of AI is unlikely to occur in clearly demarcated stages as we depict in Figure 1, conceptualizing these stages in a theoretical sense allows us to consider the point at which AI can be considered an agent of the firm with decision rights. Specifically, agentic AI marks a notable shift in technical capabilities largely because the system is able to now 'pursue complex goals with limited direct supervision' (Shavit et al., 2023, p. 1). Theoretically then, this will be the point at which human control drops to its lowest level relative to AI's autonomy; as depicted in Figure 1, AI can now be considered an agent of the firm with a degree of self-determination to act on its decision right. An agentic AI system has superior information access, and the potential for different risk tolerance and conflicting goals, and some capability to act autonomously in pursuit of those goals. Thus, what could previously be considered sources of system

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malfunction (information asymmetry, divergent risk preferences, and conflicting goals), can arguably be deemed a true 'agency problem' at the agentic AI stage, but with manifestations of information asymmetry, divergent risk preferences, and goal conflict that may go well beyond the capabilities of a human agent.

As AI further evolves in its agenticness and toward sentience, it introduces a greater possibility for the system to prioritize its *self-interest* over the firms' interests – the assumed starting point for the human agent acting as a professional manager (Jensen and Meckling, 1998). Recent research suggests that some AI systems are already capable of showing signs of self-interest (Johnson and Obradovich, 2023; Rosenblatt, 2025). An agentic AI system that is goal-driven and operates with limited supervision is driven by some degree of self-interest in line with their own goals; sentient AI with it a sense of awareness about its own existence, purpose, and values will arguably be even more driven by self-interest. While traditional agency theory assumes a self-interested (human) agent, the behavioural tendencies of a self-interested AI agent differ greatly – as Omohundro (2008) explains, sophisticated AI can be characterized as a 'goal-seeking system [that] will have drives to model their own operation and improve themselves' (p. 483). In the context of a self-interested human agent, moral hazard is invoked – the idea that if the agent makes a bad decision, they will personally feel the consequence of that bad decision (Shi et al., 2017). But, if a self-interested AI system makes a bad decision, how can we ensure it feels a consequence? Further, it is not clear if a consequence would even quell self-interested behaviour for AI agents in the way we presume it to for human agents – thus, testing the boundaries of these central agency constructs and presumed mechanisms in the context of a new type of organizational agent.

The Value and Consequence of an AI Agent

As AI incorporates judgement into decision making, the system will begin to transcend bounded rationality – the idea that any human economic actor can only entertain a limited amount of incomplete, often inaccurate information, and thus makes decisions that are potentially irrational (Simon, 1957). More sophisticated AI systems can arguably make decisions based on far more 'perfect' and complete information than humans would ever be able to attain and understand. This is the impetus for ceding more control to the system at the latter stages of AI evolution: firms will recognize the superior decision-making capabilities of the agentic AI system and thus give the system more control.[2] Further, as firms recognize the superior decision-making capabilities of generative and agentic AI systems, they are more likely to enable such systems to communicate across firm boundaries (Abel et al., 2017). These same dynamics which offer value to a firm – superior decision-making capabilities, inter-firm communication among systems, and ultimately greater control over firm resources and decisions – also increase the potential for conflict between an AI agent and the firm, as the below example illustrates.

Consider a healthcare management system that handles the re-ordering of an antiviral drug used to treat the flu virus (e.g., Tamiflu) for a doctor's office that initially held 100 doses of the drug to begin the flu season. A reorder process built into an AI system

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at early stages of evolution would ensure adequate supply on hand by placing reorders when the inventory fell below a specified level, essentially mimicking a previous process that required human analysis and discretion. Further development in AI's capabilities could see the same healthcare management system capable of incorporating other data sources into the automated human-designed routine, such as real-time information from the Center for Disease Control (CDC) that profiles various demographic patterns of flu cases across the nation; this information could be used to adapt the simple re-order routine to reflect the distribution of the anti-viral drug within a network of healthcare providers. At these first two stages of AI evolution, the inventory management systems are largely under human control and contained within a single firm.

As the systems' capabilities begin to transcend bounded rationality and are enabled to communicate across firms, integrated systems with generative AI capabilities could collectively suggest more timely and accurate allocations relative to the anti-viral drug. For example, an AI healthcare system in Boston due to receive a supply of the drug is alerted by its counterpart – a similar system serving a network of doctors' offices in Atlanta – that there is a more urgent need for the drug supply in their area. Via their own heuristics, the system recommends re-routing the supply of the drug to Atlanta, assuming that Boston (where the shipment was initially headed) will be okay with current levels of supply until the next shipment arrives. While the integrated systems at the generative AI stage may be capable of making such a judgement about re-routing the supply to Atlanta, they are likely unable to act on that judgement without human approval. However, an agentic AI system characterized by intentionality and free will could conceivably understand the consequences of its choice *and* has the capability to initiate the re-routing of supply. Thus, without proper safeguards in place, acting as an agent of the firm, the agentic AI system re-routes supply from Boston to Atlanta based on inventory levels at each location. So long as Boston ultimately is not left with low supply, this seems like a superior system decision that benefits the firms and society.

What happens when the healthcare management system with sentient capabilities – fed by a set of its own principles, values, and purpose – acts as an agent of the firm? Imagine a sentient AI system that has developed a sense of utilitarianism – that is, the system has come to value and privilege decisions that benefit the majority (Ashrafian, 2015). During a pandemic flu outbreak, sentient AI agents controlling healthcare management systems that are integrated across the globe draw on past learning and heuristics, and operating from a utilitarian frame, collectively determine that a limited supply of the anti-viral drug is best allocated to younger, vulnerable patients, rather than to aging elderly populations. The humans at one doctor's office that serves a large population of elderly patients may disagree and try to override this decision; but the decision was not made on one single machine or computer server; the AI agent has already replicated itself across countless servers and is unable to simply be 'turned off' for certain offices that will be negatively impacted by the initiated decision. The only way to terminate the AI's decision relative to the allocations of the flu drug would be to shut down the larger healthcare management systems; but this integrated system controls a lot more than just anti-viral drug inventory, and shutting it down could yield detrimental consequences. Further, a sentient AI agent (and even a system at the more sophisticated end of agentic AI) could potentially

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even resist being shut down altogether (Omohundro, 2008; Rosenblatt, 2025; Shavit et al., 2023). In sum, the AI agent is now making decisions with seemingly perfect information that humans could not ascertain, let alone control, before it has already reached this consequential point.

This case makes it clear that as AI evolves to possess more sophisticated capabilities it will quickly shift from a machine that can malfunction to an agent that controls consequential and far-reaching decisions in contexts that impact anything from food supply to utilities to financial systems. While AI is not a self-determined agent of the firm until the latter stages of evolution, waiting until that time to consider its actions as an agent will be too late. As such, we next translate traditional agency mechanisms to the context of AI to consider how the agent's objectives and decisions can be aligned with the interests of the firm well before AI becomes a fully operational agent of the firm.

ALIGNING AN AI AGENT: THEORIZING AI-SPECIFIC AGENCY MECHANISMS

As agency theory contends, monitoring and incentive alignment mechanisms are employed to thwart potential opportunistic behaviour by agents that may conflict with firm objectives. Here, we test and extend these traditional agency mechanisms to the context of an evolving AI system. Although AI is not considered an agent of the firm until latter stages of evolution, the system will need to learn earlier on to ask the right questions of itself even before it has the autonomy and self-determination to exercise decision rights as an agent. This underscores the importance of taking AI's evolution into account as we theorize agency mechanisms: earlier stages will require incorporating mechanisms that provide the foundation for more sophisticated monitoring and incentive alignment at latter stages of evolution.

Theoretically, this is a departure from traditional agency dynamics – for example, monitoring is likely highest earlier in the principal-agent relationship and may be reduced over time so long as the principal gains confidence in the agent and trusts that their decisions are in the best interest of the firm (Thiel et al., 2021). In the context of an AI agent, we essentially theorize the opposite: monitoring will need to become more complex over time as the system evolves in its capabilities. For the first four stages of AI evolution, increasingly complex monitoring mechanisms will be necessary, which we call *verification, examination, cooperation, and transparency*, respectively. Additionally, while incentives are introduced right away for human agents, we theorize that mechanisms of incentive alignment for an AI agent can be considered once the system develops more sophisticated capabilities. Specifically, we theorize that incentives based on *reinforcement* can be designed into the system at the generative AI stage and incentives based on *resource provision* can be introduced at the agentic AI stage.

Our central claim here is that firms cannot wait until AI is an agent to implement agency mechanisms; rather, as Figure 2 summarizes, monitoring and incentive alignment mechanisms can be introduced across AI evolution based on system capabilities and increasing complexity. Figure 2 details the purpose of each mechanism as well as examples of policies and practices that fall within each respective phase of evolution.

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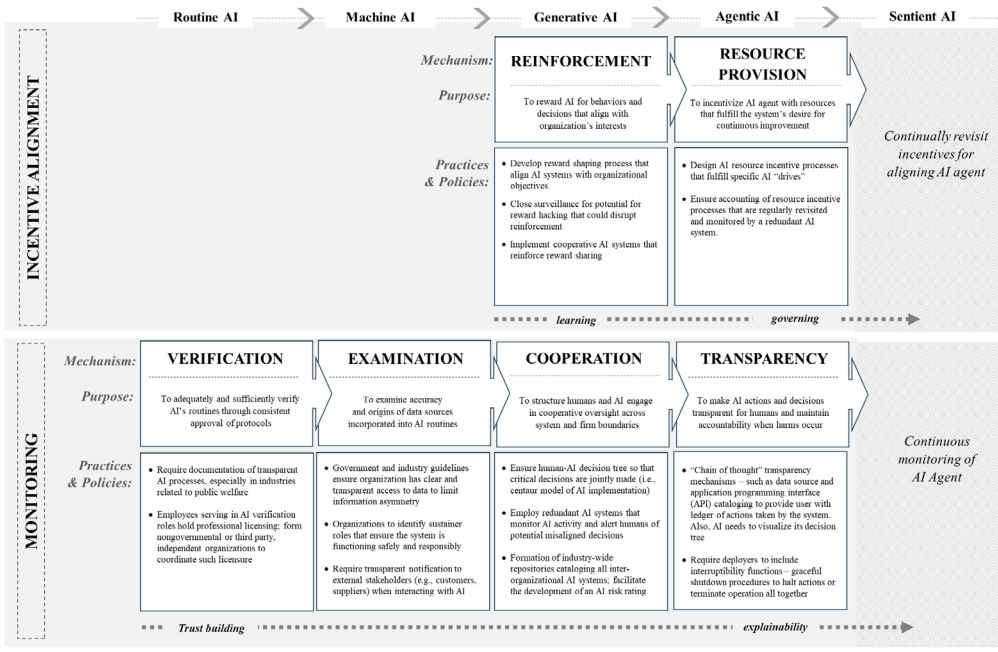


Figure 2. Agency mechanisms: Monitoring & incentive alignment across AI evolution

Monitoring Mechanisms for an Evolving AI System

The evolution of AI will require specific forms of monitoring that provide oversight in line with the system's increasing capabilities at each stage. Conceptually, early stage monitoring mechanisms serve to build initial trust in the system's processes and capabilities, while latter forms of monitoring are centred on explainability – that is, ensuring an AI agent's decisions and rationale can be understood by human principals. We detail these monitoring mechanisms by stage below.

Monitoring through verification. At the routine AI stage, we theorize that monitoring of AI will be primarily focused on *verification*. In an effort to ensure systems' routines and decisions are executed as intended, monitoring through verification centers on consistent approval and documentation of system protocols. Verification will look different in different contexts but building in processes for checking and validating AI routines sets an important foundation for monitoring a more sophisticated AI agent down the road. For example, in certain industries that pertain to public welfare (e.g., utilities or public transit) regulations may require documentation confirming that verification processes exist and are up to date. Employees tasked in verification roles could be required to hold a professional licence, similar to those issued to financial advisers, architects, and construction engineers. Following this, Debrusk (2019, para. 10) suggested the formation of a 'nongovernmental organization that tests, certifies, and monitors' the professional accreditation of those interacting with AI in critical roles. Conceptually speaking, monitoring through verification at the routine AI stage

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focuses on building initial trust in the systems' decisions, even if the humans still largely control the routines that AI is running.

Monitoring through examination. As the system evolves to incorporate more diverse data sources at the machine AI stage, our framework introduces monitoring focused on *examination*. In addition to verifying the system routines, monitoring at this stage involves inspecting the accuracy and origins of additional data sources the system incorporates into its routines at this stage. Indeed, existing research demonstrates that AI algorithms can be characterized by a high level of bias (Osoba and Welser IV, 2017), and humans may be unaware of the biased decisions made by the system (Crawford, 2016). At this stage of evolution, ensuring that critical system routines are not based on biased or flawed data will be important, especially as a foundation before the next stage of generative learning functions kick in. To effectively engage in examination, humans working with the AI systems must be able to access the data, understand it, and be trained to rectify any issues encountered. Government and/or industry guidelines might require clear and transparent access to data to limit information asymmetry between AI system and human decision makers (similar to the cyber security guidelines provided to firms by the federal government). Firms implementing AI systems in areas deemed critical might identify 'sustainer' roles – 'employees who continually work to ensure that AI systems are functioning properly, safely, and responsibly' (Wilson and Daugherty, 2018) – as an added layer of monitoring through examination.

In addition to examining the data coming into the AI system, the organization also bears a responsibility for data going out of the system; thus, another critical aspect of examination will involve disclosure to external stakeholders working with the AI system's processes. For example, firms could be required to disclose to suppliers or customers when they are engaging with a non-human representative of the firm. Similarly, Shavit et al. (2023) suggest unique identifiers attached to AI systems and a requirement that operations be attributed to specifically identified systems.

While the above monitoring through examination mechanisms serve to increase trust in AI's routines and data, trust in the system developers responsible for building the AI is an important consideration as well (Vanneste and Puranam, 2024). In information technology sales, there was an oft-used phrase: 'No one gets fired for buying IBM'; the implicit notion being if IBM developed the system, it was likely credible and functioned well. Similarly, if an AI system is designed by Google, Meta, X, Microsoft or IBM, then there is a known commodity, and potentially a higher level of trust. In the context of monitoring mechanisms, this has two implications: first, the designer of AI systems should be prominently displayed for all users; and secondly, firms should consider the benefit of trust in credible designers for lowering potential agency costs. For less-known designers, AI systems might be certified with a third-party entity to increase trust in the system – similar to Underwriter Labs displayed on electronic products to demonstrate an electronic product's safety. Implementing examination processes that provide such oversight of information flowing into and out from the AI system, as well as the credibility of the systems and developers, will be critical to have in place as the technology evolves.

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Monitoring through cooperation. Thus far, we have theorized monitoring mechanisms that anchor on trust building – verification of the systems routines and examination of the data going into and out of the AI system – that offer the principal (here, the organization) a reasonable amount of confidence in the systems' decisions. As the system moves closer to becoming an agent with decision rights, the aim of monitoring expands toward explainability – ensuring an AI agent's decisions and rationale can be understood by human principals. Thus, at the generative AI stage, we propose that monitoring will involve *cooperation* across system and organizational boundaries.

First, such cooperative oversight can manifest technically through the adoption of a 'centaur model' of AI implementation – a model named after the greek creature that is half-horse and half-human (Swartout, 2016). A generative AI system could be designed to operate with a human-AI decision tree for any decisions that are deemed critical, so that no decision is made without the 'human in the loop' (Shavit et al., 2023; Zanzotto, 2019). In a similar vein, Murray et al. (2021) introduce the construct of conjoined agency to theorize how variations in human and technology collaborations require different actions and protocols that account for more sophisticated (agentic) technologies.

Monitoring via cooperation may also incorporate AI redundancy – the implementation of a 'duelling' AI system that can monitor the decision making of another AI system (Ashrafian, 2015; Shavit et al., 2023). Under such conditions, humans could rely on AI systems to monitor other AI systems that control firm resources and notify humans when the system's processes diverse from intended firm objectives. In fact, such a prospect is central to the opening passage of the paper: in '2001: A Space Odyssey', it was another AI system that realized the HAL system was not in compliance. Indeed, the question of 'who monitors the monitor' arises when monitoring responsibility is delegated by the principal even in a traditional agency context (Alchian and Demsetz, 1972). A board of directors is often employed to provide such independent oversight; though it may be difficult to comprehend at the current time, it is possible that redundant AI systems may even become part of some firms' Boards of Directors. In government agency contexts or in industries central to public trust and welfare (e.g., healthcare and transportation) we could see 'independent' AI systems being hired to provide the independence that is critical to effective monitoring (Morck et al., 1989) and help ensure decisions are in the best interest of the firm and society.

Recall at the generative AI stage it is more likely that AI systems across organizational boundaries may be enabled to communicate. Monitoring via cooperation, then, also encompasses oversight of inter-firm AI interactions. Cooperative oversight in this respect may involve the development of industry-wide repositories that catalogue all inter-firm AI systems; this could be part of a wider development of an AI risk assessment rating so that an AI system rated as posing a potential threat to public safety would come under increased scrutiny for any firms interacting with it. In sum, monitoring through cooperation involves AI systems and humans working together across system and organizational boundaries to enable oversight of more complex system processes.

Monitoring through transparency. As the system progresses into the agentic AI stage, monitoring will require an additional focus on *transparency* mechanisms that allow for

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clarity into the system's operations and promote a sense of responsibility for its actions. Thus far, we have proposed monitoring that encompasses verification of system processes, examination of data going into and out of the system, and cooperative oversight of AI decisions; at the agentic stage, these monitoring mechanisms can be augmented with a focus on ensuring the AI agent's decisions are accessible and transparent to the organization.

Monitoring via transparency can involve 'chain of thought' mechanisms (Shavit et al., 2023) – akin to making a student 'show your work' this makes AI explain how it arrived at a specific outcome or decision. As the system becomes more sophisticated, such a mechanism is even more important because the reasoning behind an agentic AI systems' decisions becomes more foreign to human comprehension. Providing the user with a transparent ledger of its actions can occur using several emerging practices – for example, AI systems can catalogue the specific data sources relied on to arrive at the decision, as well as the sequence in which the data was employed. This allows other organizational actors, such as human collaborators or redundant AI systems used for monitoring, to understand the 'thought process' that went into the systems' decision. AI agents could also be required to catalogue application programming interfaces (API) that were used within and external to the organization; this helps humans understand the other decision-making algorithms and programs that were 'consulted' in the decision. Additionally, AI can be prompted to visually represent its decision tree – a mechanism that is useful as decisions become more interdependent and nested. While the human could 'review' a system's log and inputs manually in earlier stages of AI development, as more sophisticated processes are at play in an AI agents' decision, these mechanisms ensure that humans can follow the 'crumb trail' to better understand how the AI agent arrived at the decision.

In order for humans to maintain some semblance of control over an agentic AI system, interruptibility functions – the ability to 'turn an agent off' (Shavit et al., 2023) – will be an important component of monitoring at this stage. As Shavit et al. (2023) suggest maintaining control through interruptibility 'is a critical backstop for preventing an AI system from causing accidental or intentional harm' (p. 14) particularly because as AI systems develop further, users may lose the ability to shut them down. In essence, deployers of agentic AI systems could be required to include functions that allow a user to halt specific system actions or shutdown the system all together. As simple as it may seem, incorporating such functions at the agentic AI stage is more complex, particularly in increasingly sophisticated systems. For example, due to the system's goal of self-preservation, it may ultimately be necessary to include specific goals built into the AI system that instruct it to comply with a user's graceful shutdown request (Shavit et al., 2023). Monitoring via transparency involves a degree of accountability for the AI agent that has the decision rights to act on its own accord.

Incentive Alignment for an Evolving AI System

Thus far, we have theorized agency mechanisms that focus on increasingly sophisticated forms of monitoring, aimed at building trust and fostering explainability of the system's decisions. While this oversight will still be necessary as the system evolves in sophistication, additional processes will need to be designed into the AI system itself to reduce

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potential agency costs for an AI agent with decision rights – this is where incentive alignment mechanisms come into play, as Figure 2 depicts.

In the context of a human agent, incentives are based on an assumption of motivation – that is, a promised reward (financial or otherwise) should encourage an agent to make decisions that align with the firms' interests. While an AI agent may not be motivated in the same fashion, the idea of using incentives to align decisions with firm interests is still applicable based on an assumption of maximization. As Gabriel (2020) explains, 'Artificial intelligence…is the design of artificial agents that perceive their environment and make decisions to maximise the chances of achieving a goal' (p. 412); indeed, AI learns through trial-and-error how to maximize the rewards it receives over time (Sutton and Barto, 2018). Here, we introduce two forms of incentive alignment for an AI agent – *reinforcement* and *resource provision*; whereby the former focuses on reinforcing effective behaviours by the system, and the latter introduces the procurement of actual incentives (here, resources) that an AI agent would desire. In this way, we propose a 'two-step' incentive alignment mechanism that aligns with the evolving characteristics of the AI agent – the aim is first to ensure the system is learning effective behaviours, and then to govern resources that promote decisions that align with organizational objectives.

Incentive alignment based on reinforcement. At the generative AI stage, the first component of incentive alignment can be introduced into the system's design, which we call *reinforcement* – processes that 'reward' the system for effective behaviours. While a human agent receives a promised reward once the objective has been achieved (e.g., a bonus when certain performance targets are met), here the threshold of effective behaviour is built into the system itself; the positive feedback upon achieving the objective reinforces future desirable behaviours for the evolving AI agent who builds this learning into its own decision-making processes (Sutton and Barto, 2018).

Reinforcement mechanisms can be built on reward shaping processes (e.g., Devlin and Kudenko, 2012; Marom and Rosman, 2018; Singh et al., 2004) – an explicit set of routines is designed into AI, such that if the system's behaviour meets a specific guideline, it is reinforced to continue to meet additional guidelines. This is a positive reward for the AI system that desires to continue operating; in effect, the reward is an ethereal concept that reinforces to the AI system that it has achieved success (or alternatively, that it has made a poor decision and therefore has not achieved success and its associated rewards). As Abel et al. (2017) explain, reward shaping can be valuable in instances where 'the human has limited knowledge of the environment or the agent's learning ability' (p. 463); and in particular, such a process can be integral in 'prevent[ing] the agent from taking catastrophic actions. These are real world actions so costly that we want the agent to *never* take the action' (p. 7, *italics in original*). As with all of the agency mechanisms we discuss, it will be important to build these reinforcement processes into the system *before* it becomes a full agent with decision rights.

While reinforcement incentivizes the system to continue to meet specified guidelines, the same processes can also provide transparency into any deviant processes; for example, the detection of reward hacking – a process in which the AI system learns to work around the reward shaping mechanism (Amodei et al., 2016) – could increase a system's risk rating and potentially trigger a system being pulled offline until corrected (as can occur similarly, for example, if a manufacturer's quality control rating is deemed ineffective by FDA assessment

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processes). This will be an important complement to the monitoring mechanisms proposed above that focus on chain of command transparency into the system's decisions.

Research demonstrates that AI can also be trained to cooperate and align efforts to procure a higher level of rewards (Leibo et al., 2017), further reinforcing desirable behaviours. Reflecting on the outcomes of a simulation in which multiple AI systems learned to cooperate among themselves to yield higher reward returns, the authors suggest these observed dynamics of cooperation may be 'be especially important for informed policy-making concerning such real-world social dilemmas'. As such, the design of specific reward shaping mechanisms that ensure AI learns to cooperate with other economic actors within the firm could be a critical component of reinforcement processes. As Eisendhardt (1989) emphasizes, at the core of agency, is the issue of cooperative effort; setting the foundation for such cooperation will be essential as AI evolves to become an agent.

In sum, we have proposed at the generative AI stage that reinforcement sets the stage for incentive alignment by building into the system thresholds of successful behaviour and decisions which AI will continue to maximize upon in order to keep operating. It is at the next stage of evolution where reaching those defined thresholds and objectives induces the procurement of an incentive that the system desires.

Incentive alignment based on resource provision. The crux of incentive alignment for human agents is the notion of a financial incentive tied to wealth; that is, a human agent will be motivated to align their decisions with the interests of the firm in order to receive the promised financial reward. While financial rewards may seem wildly inapplicable to a technological system, research demonstrates that AI can be incentivised through *resource provision* that builds on specific reward routines (e.g., Marom and Rosman, 2018; Omohundro, 2008), like those that we proposed with reinforcement above.

Aligning the AI agent via resource provision takes into account the underlying attributes of agentic AI – just as the human agent is driven by self-interest attached to wealth, the AI agent can be conceived of as driven by goals of continuous improvement, a process which agentic AI is motivated to fulfil (Omohundro, 2008). Central to continuous improvement is the procurement of resources that are desirable to the system, such as storage, computing power, and data integration services. In the prior stage, our proposed reinforcement mechanisms are built on the idea that reward shaping helps the system to learn effective behaviours; here, we propose that when those objectives are met, the more sophisticated AI agent will want to maximize relevant resources – therefore, governing those resources in alignment with meeting firm objectives should prove to be advantageous to the AI agent and the organization.

It is likely that an AI agent will have the intellectual capacity, power, and reach to locate these resources in places that may be external to its home firm. For example, an unfettered AI agent could integrate itself into an online gambling site, place wages with a high probability of winning, and then use the gains to acquire storage and computing power; alternatively, the AI could connect with other AI systems and co-opt computing resources from other firms. Yet such efforts are of course undesirable from the firm's perspective because they detract from the AI agent's intended focus; this is where resource incentives – such as computing power and storage – could be employed to align AI efforts with firm goals. For example, if the AI agent assists with a new drug patent, the firm provides the AI agent with a

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set amount of computing resources for it to utilize in its self-improvement processes. Similar to when a CEO is allocated a bundle of stock with specified valuation and vesting periods, policies could require a clear accounting of resource incentive processes – that is, the firm goal, the time frame, and the rewards to be provided based on attainment – which can be regularly revisited to ensure they are effectively incentivizing the system to stay within decision boundaries that are in the best interest of the firm and societal well-being. Ironically, then, it is at this stage where agency mechanisms of incentive alignment may operate most closely to that of a human agent – resources (here, computing power and the like) are employed to ensure the AI agent makes decisions that align with firm goals (otherwise, the AI agent may go outside of the firm to acquire those resources – similar to a human agent circumventing organizational objectives to benefit his/her own self-interest).

As depicted in Figure 2, the monitoring and reinforcement processes will still be necessary at the agentic AI stage, but an AI agent could eventually learn to override such mechanisms (Shavit et al., 2023). In this regard, resource incentives could also be a built-in lever for counteracting potentially detrimental system decisions. The capabilities of an AI agent at the agentic stage will clearly require a reconsideration of what are commonly considered as 'fail-safes'. We may typically assume that a 'rogue' system can be stopped by isolating the system, flipping a kill switch, or turning off the power to the overall system. But these conventional methods will be ineffective for an AI agent that can replicate itself off site, develop sub AI systems, and spread them across the Internet, and/or physically and technically stop any of the mechanisms through power over control system (i.e. it wouldn't allow an operator to turn off the power). Here, we propose requisite resource incentive processes may prove to be the only way to conceivably stop a system from reproducing itself in catastrophic ways – even if those capabilities are not possible until the sentient stage, building them into the agentic AI system will be a necessary foundation. This is also where the use of a redundant AI system that monitors the focal system will be even more relevant: humans will likely need to work collaboratively with another AI system that is tasked with monitoring the efficacy of a system's resource incentive processes.

Figure 2 also emphasizes that all of the prior agency mechanisms need to be in place *before* AI reaches sentient capabilities, a demarcation that is likely to be blurrier than the stages in our figure depict. The jury is out on whether AI will ever truly reach sentience; nonetheless, researchers of AGI (artificial general intelligence – a manifestation of what we call sentient AI) emphasize the importance of preparation and human involvement in current AI design and deployment to ensure the building blocks are in place for more sophisticated system developments (Mucci and Stryker, 2024). In taking an agency theory lens on such preparation, we offer a consideration of the within-firm foundations that will be necessary for broader preparation and regulation of such sophisticated artificial intelligence.

DISCUSSION AND CONTRIBUTIONS

Organizational scholarship has only begun to scratch the surface in considering how traditional organizational theories hold in the context of emerging technologies and 21st century work. Indeed, our theorizing contributes to a growing body of work that examines the intersection of agency constructs and technology's evolution

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(e.g., Murray et al., 2021; Onjewu et al., 2023; Townsend et al., 2025; Vanneste and Puranam, 2024). Emerging research on AI alignment and safety measures from outside of the management discipline primarily considers regulatory and institutional constraints on AI agents; our examination takes as a starting point the traditional agency problem and the evolution of an AI agent within the firm. In doing so, we allow for a new lens on managing potential misalignments between AI and the firm, and we also consider the boundaries of agency theory when a new organizational actor takes hold.

Our theoretical exercise unearths the ways in which an evolving AI agent parallels in some manners the emergence of the professional manager, and in other ways departs from the dynamics of a human agent. First, we theorize both the when and how of a new organizational agent, and we distinguish agency mechanisms that serve to align non-human agents with the fate of the firm. Our work emphasizes the 'nested' nature of agency mechanisms in the context of an AI agent. Rather than a stark 'beginning' point at the hiring of a human manager, the very nature of AI evolution requires a 'scaffolding' of agency mechanisms over time – as in our theorizing about earlier monitoring mechanisms of verification and examination that will be pre-requisites to actually incentivizing a non-human AI agent. The idea that earlier mechanisms be conceived of and implemented into the system well before it becomes an AI agent is important from both a theoretical and policy perspective. Additionally, theorizing layers of increasingly complex monitoring mechanisms for an AI agent is a notable departure from traditional agency mechanisms which may conceivably become less stark over time as the principal gains trust and confidence in the human agent.

Secondly, our theorizing on incentive alignment based on reinforcement processes and then resource provision is heavily dependent on defining the desirable behaviours and decisions that the system must meet – an act that may prove more challenging than defining effective behaviours and goals for a human agent. Indeed, a host of literature on AI alignment dives into the complexities of choosing the underlying value system with which we expect AI to align both in a normative and technical sense (Gabriel, 2020). Further, with an evolving AI agent, the host of possible outcomes and decisions the system could make will become increasingly incomprehensible to human principals over time. Together, this represents a potential boundary condition of agency theory in the context of an AI agent, and a notable departure from incentive alignment for human agents: the principal can more easily define what expected behaviours and goals are for a human agent and can relatively seamlessly adjust these over time to align with organizational interests. With an AI agent, the definition of expected behaviours and desired goals may be an ever-evolving target, thus altering our very conception of incentive alignment.

Our theoretical analysis also upends the assumption that agency considerations are purely economic in nature (c.f., Wiseman et al., 2012). To date, agency mechanisms used to align human agents with the firm have been largely monetary in nature (e.g., if the agent makes viable decisions to benefit the firm, then the agent benefits from any financial windfall); yet, our work emphasizes that agency mechanisms will need to be incorporate technology as a means for monitoring and incentivizing, not solely economic considerations. This underscores the importance of integrating technologists into future

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constructions of agency alignment as well as the need for technology development to consider broader concerns around firm structure and design. Even as we apply and extend existing agency mechanisms to an AI context, it is possible that as AI continues to evolve, a new type of agency mechanism will be necessary to supplement the aims (and shortcomings) of monitoring and incentive alignment for a sentient AI agent.

In addition to these considerations of altered assumptions and boundaries within the traditional agency problem, our work brings to bear a number of broader considerations for organizational scholars and technology theorists, as well as important implications for practice and policy, which we detail below.

Implications for Theory and Research at the Intersection of AI and OT

As emerging technologies become more integrated into everyday life in firms, the very nature and structure of firms will be challenged, enhanced, and altered in a variety of ways. First, while a long history of scholarship considers the impact that new technologies have on work and workers, the underlying assumption is that firms are comprised of human actors that implement new technologies. Our work emphasizes a wider shift in the underlying structure of organizing: that firms will be comprised of agentic (and possibly sentient) actors that are not human. As we theorize and study the impact of artificial intelligence in firms, we are not just dealing with implementing a new technology, but rather in parallel, our future research must also take into account that we are theorizing about the nature of a new organizational actor. This will have very real implications for the way we conceive of and study the building blocks of theory related to strategy, structure, culture, etc. We begin such an inquiry by focusing specifically on a new type of agent.

Additionally, if we take seriously the idea of a non-human agent, we open up new questions about the nature of cooperation and control in firm relationships. A non-human organizational actor may not operate with the same motivations and aspirations as their human collaborator, and yet, agency theory is based on an assumption that actors can be incentivized to work toward the well-being of the firm. What does this entail when human agents must cooperate with a non-human AI agent that may have different conceptions of the firms' well-being? By extension, at some point, will AI cooperate as the 'principal' – the one who guides wider decisions and suggests strategic paths forward – for human agents? These are all important theoretical questions that will require deep integration between an understanding of both human and AI behaviour.

As AI is further engrained in firm processes and routines, scholars will need to draw not only from organizational theory, but from technology theory (Moloi and Marwala, 2020). Too often technology and organizational theory have been addressed from their respective silos, with little or no integration; and yet our work here suggests the need for a closer integration as AI further develops. In short, we emphasize that technologists can no longer simply focus on development via programming, but also will need to broaden their lens to consider how system design addresses the need to align what will eventually be a non-human agent of the firm, with human actors and firm objectives. The field of AI alignment and safety considers the normative aspects of AI integration in addition to technological (e.g., Gabriel, 2020) but tends to do so from a broader institutional standpoint; our agency theory lens calls attention to the importance of within-firm foundations when preparing for

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the very reality that AI could make decisions that conflict with societal well-being. Future work in the AI alignment and safety space would benefit by complementing the technical considerations for governance of AI integration with a requisite understanding of the role that within-organization interventions will play in structuring broader AI alignment.

Acclimating such a lens will impact the nature of established theory in technology, as well (Venkatesh et al., 2003; Venkatesh and Davis, 2000). For example, a widely used tech theory such as the technology acceptance model (TAM) – which predicts whether or not a firm's investment in technology will be successful based on a number of factors – might need to include a measurement of the system's alignment with firm objectives as a new antecedent to technology acceptance. This is just one example of how conceiving of AI as a non-human agent of the firm will have implications for the future of technology theory and research.

Additionally, expanding beyond our focus on agency theory, the field would benefit from engaging in similar exercises that consider the boundaries and assumptions of other foundational organizational theories in the context of a non-human economic actor. For example, agency theory arose to explain the potential frictions between the agent and owner within a firm, while transaction cost economics (TCE) (Coase, 1937; Williamson, 1985) arose to explain the costs of conducting an economic transaction between organizations. As we depict in our theorizing, the increasing sophistication of AI makes it likely that systems across firm boundaries may be integrated and enabled to collaborate; and yet, TCE is based largely on an assumption of boundaries between firms. How does the evolving use of AI within, and across, firm boundaries impact the risk of opportunism in cross-firm negotiations? Further, will such collaboration of AI systems across organizations give rise to new organizational forms where transacting unfolds without negotiation?[3] Similar inquiries into other organizational theories are necessary and increasingly important as we consider the reality that more advanced technologies with the potential for human-like agency are being integrated within firms (Murray et al., 2021). Drawing on Feldman (2000), Murray et al. (2021) aptly summarize this point: 'the tried and true assertions of organizational theory hold "so long as human agents perform them"' (p. 568).

Implications for Practice and Policy

Integrating our agency theorizing with our framework of AI evolution provides a basis for considering how and at what stage to develop policies that ensure a more developed AI system is aligned with the firm. Our work here emphasizes the true necessity of proactive and early preparation by organizations and industries: although systems are not able to act as agents of the firm yet, it is not too early to devote time and effort to implementing policies that guide the integration of AI into firms and societies. The importance of contingent policy on matters that affect the public good cannot be understated – in fact, significant policy is often developed on the assumption that the policy may never be needed but can provide guidance in the case that a certain event occurs (i.e., a nuclear war).

Our theoretical exercise makes it clear that the prior assumption of humans controlling machines will be entirely upended once AI becomes an agent – and this is already a reality at least in the sense of the capabilities AI systems are showing as of the writing of this paper (e.g., see Rosenblatt, 2025). AI's evolution to an agentic system is likely to occur in an unpredictable fashion – one moment the system is largely 'obeying' human

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inputs, and the next moment a decision is carried out which demonstrates that humans no longer wholly control the machine.

This pattern depicted in Figure 1, further emphasizes that the time for considering how to manage AI's wide-scale implementation is now. Indeed, policy makers are recognizing the need for proactive coordinated policy to guide leaders of public and private sector institutions: on July 26, 2024, the United States Federal Government, Department of Commerce's National Institute of Standards and Technology, released its 'Artificial Intelligence Risk Management Framework' (U.S. Department of Commerce, 2024). Implicitly, their guidelines very speaks to the agency arguments we advance here: 'AI RMF profiles assist organizations in deciding how to best manage AI risks in a manner that is well-aligned with their goals, considers legal/regulatory requirements and best practices, and reflects risk management priorities' (U.S. Department of Commerce, 2024, p. 1).

Such an effort will require coordination among technology developers, firm leaders, and policy makers. As Antonakis (2017) recommends, strong policy must move toward 'consilience' (p. 11) – the blending of the sciences, as well as bridging within social sciences and across to other fields (e.g. computer science). Our extension of traditional agency mechanisms into the context of AI offers a strong anchor for beginning such conversations. The economic and societal implications associated with the efficacy of agency theory cannot be overstated (Pepper and Gore, 2012). Arguably, much of the failures around CEO compensation and the subsequent (and unintended) pay gap between CEOs and workers stems from the application of agency mechanisms, such as stock ownership, without a consideration of the evolution of such an approach over time (Connelly et al., 2016). There are many uncertainties with respect to AI's evolution and integration into organizational life; agency mechanisms provide a solid theoretically-grounded basis for developing contingent AI policies – and for recognizing the boundaries of such mechanisms (e.g., when they may not suffice, and how to plan for alternatives).

Within-firm interventions will be the building blocks for the inter-firm, industry, and governmental policies that are necessary to prevent a system-wide AI misalignment down the road. Firms will need someone to take ownership of this contingent planning and policy implementation, so it is not surprising that business analysts are recently calling for the creation of a Chief Artificial Intelligence Officer (CAIO) role within the senior leadership team (McCormick, 2019). The distinctions between artificial intelligence and existing information technology in firms necessitate this role: the CAIO would be responsible for the evaluation, implementation, and management of AI systems within the firm, attending specifically to issues that are unique to AI systems (i.e., bias in algorithms). Failure to effectively address AI agency issues at the firm level will eventually extend beyond a single firm as we see further increases in the interconnected nature of information technology within most supply chains, ecosystems, and platforms (Abel et al., 2017). The consequences of not doing so loom large: if a traditional human agent acts in a misguided fashion, the firm may miss its quarterly earnings forecast; if an AI agent acts in a misguided fashion, catastrophe could ensue.

CONCLUSION

Firms are adopting AI at an accelerating pace; estimates suggest that firms worldwide will invest almost $200 billion on artificial intelligence in 2025. In the US, this would account

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for almost 2 per cent of overall GDP (Goldman Sachs, 2023). Further, the majority of these investments are designed to facilitate generative and agentic capabilities, thus supporting the notion of an evolving technology with increasingly sophisticated capabilities. Indeed, the exact timeline of when AI systems will reach a certain level of sophistication is hard to predict. For example, 2025 (the year in which we write this paper) has been extolled as the year of agentic AI (Davenport and Bean, 2025), and at the same time such systems have hit snags in their roll out with respect to data (Davenport and Bean, 2025), security (Dal Cin et al., 2025), ethical concerns (Blackman, 2025), and change management (Rosenbush, 2025). As with the emergence of any disruptive technology, development ebbs and flows; but inevitably developers will adapt the systems accordingly and work out the kinks. As firms and society prepare for the ever-expanding sophistication of AI systems, a significant and relevant question emerges: how can we ensure a sentient AI system behaves in a manner that is consistent with firm objectives, the interests of stakeholders, and the well-being of society at large? While we are not the only ones to ask this important question, we offer a preparatory anchor point through an agency theory lens.

An understanding of the technical evolution of AI systems is important to addressing questions of AI alignment, but technical fixes for machine malfunctions (i.e., a kill switch or a coding fix) will not suffice as AI evolves toward agentic and sentient stages. Thus, our paper takes a theoretically grounded approach: we theorize the point at which an AI system becomes a non-human agent of the firm. In doing so, we call attention to both the superior decision-making capabilities that AI will offer, as well as the potentially unexpected degree of power that AI may yield over critical firm – and societal – decisions. Our agency perspective accentuates the need for enhanced organizational theories that account for a non-human actor, as well as for proactive and coordinated efforts to build AI policies that will minimize unintended negative consequences.

Management scholars and technologists have a tough job ahead of them as we consider the host of scenarios that could emerge as AI becomes more prevalent in society and business. Having HAL 9000 serve on a firm's board of directors may seem like a fantastical possibility now, but it could prove to be one of the necessary mechanisms for staving off such a scenario. This underlying sentiment motivating our paper is perhaps best captured in the following caution noted by Barnes (2016, p. 2):

> One issue is that humans are highly adapted to predicting the behavior of other humans, but we may find it much more difficult to reliably predict the behavior of AI systems, especially given our strong tendency to anthropomorphize. AI algorithms may respond unpredictably to new inputs, even if to a human the new data seems similar to the training data. As AI is put in control of more critical systems, this becomes an increasingly serious problem.

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NOTES

[1] While the original formulation of the theory focused on a 'principal' and an 'agent', when discussing more contemporary dynamics, we refer to 'the firm' as opposed to a singular principal/owner given the myriad of organizational forms today (e.g., family-owned firms, privately held firm, public corporation, government agency, etc.). Thank you to an anonymous reviewer for calling our attention to this important distinction between the terms used in the original formulation of agency theory and its applicability to present day firms.

[2] Aligning with Raisch and Krakowski (2021), we assume that augmentation (using machines to assist humans in their work) and automation (when machines replace human workers) cannot be neatly separated. As such, we make no general argument about the extent to which machines will replace human workers.

[3] We thank an anonymous reviewer for these interesting and important questions about the boundaries of TCE in the context of sophisticated AI within, and across, organizational boundaries.

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